

ROHAN KRISHNAN

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EDUCATION

M.S.E. in Data Science, University of Pennsylvania, Philadelphia, PA Expected 2027

- Coursework: Big Data Analytics, Applied Machine Learning, Python Programming for Data Science & AI

Hons. B.S. in Economics & B.S.B.A. in Business Information Systems, University of Pittsburgh, Pittsburgh, PA 2020 – 2024

- Minor in Applied Statistics & Certificate in Business Analytics
- Honors/Awards: Summa Cum Laude, \$40,000 Merit Scholarship, \$4,000 Honors Research Fellowship, \$3000 DSG Analytics Case Competition 1st Prize, PLUS 3 International Business Fellow, Haller Global Honors Fellow

WORK EXPERIENCE

Business Analyst, Penn Graduate Consulting Club / AI Maternal Health Startup, Philadelphia, PA Mar 2026 – Present

- Developing value-based pricing models based on empirical health research and Philadelphia market data using Excel for an AI-powered early pregnancy care platform.
- Using Python to analyze government census and health datasets to validate value proposition alignment for B2B and payer-focused segments versus traditional ED visit pathways.
- Conducting Monte Carlo simulations in Python to better simulate uncertain and “low data” problem space and generate ranges of potential value delivery for potential payer and health system clients.

Freelance Data & Automation Consultant, Self-Employed / Local Healthcare Clinic, Mechanicsburg, PA Jul 2024 – Nov 2024

- Engineered a HIPAA-compliant Python ETL pipeline to clean and validate billing data for a practice with a \$400K billing backlog, reducing manual data preparation time by 70%.
- Built an MVP of an audio-to-structured-data transcription tool in Python for physician notes, cutting redundant data entry time by 90% and accelerating revenue cycle resolution by 45+ days.

Technology Co-Lead, Pitt Business Advanced Analytics Consulting / Metal Manufacturer, Pittsburgh, PA Jan 2023 – Apr 2023

- Built an ensemble raw metal price forecasting model (ARIMA, prophet, random forest) achieving 85% F1 accuracy on price movement classification and MAE within \$3 for spot price prediction, enabling an estimated \$400K+ in procurement cost savings on \$5M+ in zinc purchasing decisions for a national metals manufacturer.
- Delivered an R Shiny dashboard visualizing model predictions and procurement metrics to client stakeholders while co-leading full technical development across a 12-member consulting team.

PROJECTS

Racquet Semantic Search, Hybrid semantic search engine for tennis racquet discovery Mar 2026 – Present

- Building a hybrid semantic search engine over 320+ tennis racquets combining BM25 lexical retrieval with dense embeddings; used LLM-distilled product descriptions to improve embedding quality over raw marketing copy.
- Designed and implemented end-to-end scraping and cleaning pipeline across 11 brand pages, engineering 15+ structured features from free-text spec fields via regex parsing.
- Prototyped a conversational search UI using AI-assisted front-end development to validate UX and card design before backend implementation.

Flight Delay Triage System, ML-based operational risk ranking for incoming PHL flights Feb 2026 – Mar 2026

- Scraped and merged BTS flight delay records with destination weather data to build a delay prediction model for incoming PHL flights; reasoned through eval metric selection and determined raw classification was the wrong output for the business problem.
- Pivoted to using predicted probabilities from the best-performing model to rank incoming flights by delay risk, reframing the problem as an operational triage tool for directing airport personnel attention.
- Refactored and extended a prior class project into a production-quality repository with structured methodology, clean documentation, and a complete README.

Datashelf, Open-source Python library for dataset version control Jan 2026 – Mar 2026

- Designed and published a lightweight Python library for dataset version control, implementing a SHA-256 hash-based deduplication, automatic file parsing and conversion, and YAML metadata tracking.
- Built a CLI and Python API modeled on git workflows, enabling data scientists to save, load, and audit dataset versions across analysis stages without external infrastructure.
- Structured the project as an installable package with versioned releases and a public PyPI distribution.

SKILLS

Python (pandas, scikit-learn, NumPy, FastAPI, matplotlib, BeautifulSoup4, Streamlit, PySpark), R, SQL, Tableau, Git